Adaptive Retrieval for Evidence-Grounded Answers

Synthetic research fixture, not a real publication.

Abstract

This self-authored fixture describes an imaginary comparison between a fixed retrieval baseline and an adaptive evidence selection method. Its purpose is to exercise reading, text selection, citation navigation, and research notes in the Read-paper workbench.

All methods, data, and results are synthetic. They must not be used as claims about a real system or publication. The document includes ordinary selectable text and stable page boundaries for reproducible product tests.

1. Research question

When does adding a retrieval stage improve the factual quality of an answer? We compare answers grounded in a relevant evidence collection with answers grounded in a collection containing unrelated material.

A useful comparison records the question, the available sources, the selected evidence, and the resulting answer. The answer can then be checked against the same source excerpts that were available at generation time.

2. Evidence collection

The imaginary corpus contains short notes about a single research question. Each note has an identifier, a page number, and an excerpt. We retain these fields together because a quote without its source cannot support a reviewable conclusion.

3. Synthetic procedure

The fixed baseline always selects two notes. The adaptive method first checks whether a candidate note is relevant to the question, then chooses up to two relevant notes. Both methods receive the same synthetic corpus and the same answer length limit.

A reviewer marks each statement as supported, contradicted, or absent from the selected notes. Supported statements are counted only when the source excerpt can be recovered by its document identifier and PDF page index.

4. Imaginary results

In the relevant-corpus condition, the adaptive method produces more supported statements. In the unrelated-corpus condition, selecting additional notes does not create useful evidence. These are invented observations for software verification.

Relevance is a condition of the example, not a universal property of retrieval. The test deliberately preserves this qualification so that cards and conclusions can represent both the claim and its limits.

5. Reproducibility notes

A reliable answer should preserve a traceable link to its supporting evidence.

The fixture generator fixes metadata dates, typography, page size, and content. Re-running the generator should create the same bytes. The paper uses a single text column and three PDF pages.

Text extraction and rendering are separate capabilities. This fixture has a text layer, while the accompanying scanned research note contains a raster image and no selectable page text.

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6. Main finding

Retrieval improves factual accuracy when the evidence corpus is relevant.

This exact sentence is the selection target for the end-to-end research workflow. It is intentionally located on PDF page 3, whose internal page index is 2. It should remain selectable after zooming or reopening the document.

7. Conditions and limits

The finding is conditional on source relevance and recoverable citations. A fluent answer may still contain unsupported details. Reviewers should inspect the original excerpt rather than treating the presence of a citation as proof of correctness.

The fixture makes no claim about real benchmark performance. It exists to demonstrate evidence cards, shared card reuse, board organization, and a conclusion export that includes bibliography details, PDF page numbers, and quotations.

8. Conclusion

A research note is easier to verify when its claims preserve the conditions under which they were observed. Explicit source excerpts let the reader revisit the same evidence while writing a conclusion.

Suggested synthetic citation: Read-paper fixture authors (2026). Adaptive Retrieval for Evidence-Grounded Answers. Self-authored software test material; not a real publication.